Semester Project: Domain Discovery, Recommendation, and Graph Intelligence

Milestone: Feature Engineering and Dimensionality Reduction (Week 5)

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1. Overview

This milestone covers the construction of the three feature matrices used as input to all downstream models (clustering, recommendation, graph analysis), and the dimensionality reduction analysis used to select compression parameters.

Code: src/feature_engineering.py | Artifacts: artifacts/ | Figures: src/generate_figures.py

2. Dataset Snapshot

- Processed dataset: data/processed/steam_v1.parquet - 480,025 interaction records
- Games with tag metadata (clustering): 69,943 titles
- Games in feature matrices: 87,806 titles (full catalog)

3. Feature Matrices Built

Matrix	Shape	Type	Content
X_numeric	87,806 x 26	Dense	Numerical catalog fields. StandardScaler applied.
X_categorical	87,806 x 5,000	Sparse CSR	Top-5,000 genre/category tokens. Binary encoding.
X_text	87,806 x 5,000	Sparse CSR	TF-IDF on name + description + tags (1-2 grams).

Numeric columns (26)

Required age, price, DLC count, Metacritic score, achievements, recommendations, user score, score rank, positive/negative counts, estimated owners, average/median playtime (forever and 2-week windows), peak CCU, pct_pos (total/recent), num_reviews (total/recent), Windows/Mac/Linux flags, release year, release month, release age in days.

4. Dimensionality Reduction Analysis

PCA on X_numeric

- Components tested: up to 26 (full rank).
- Result: k=12 components capture >= 90% of variance.
- 12 orthogonal directions cover price tiers, engagement levels, platform availability, recency.

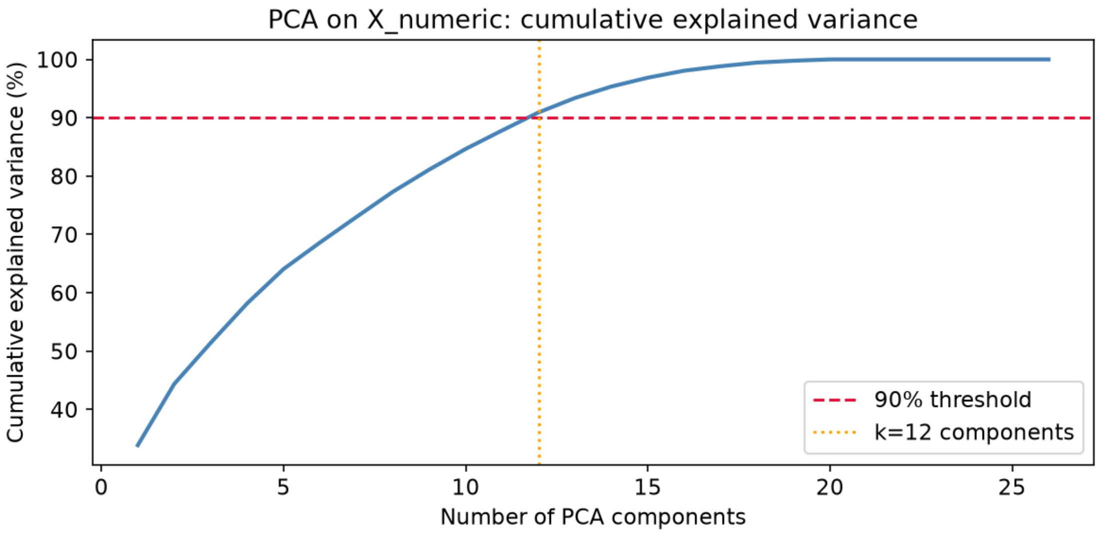

Figure: PCA on X_numeric: cumulative explained variance. Red dashed = 90% threshold; orange dotted = k=12.

TruncatedSVD on X_text

- Components tested: up to 300.
- Result: k=201 components capture >= 50% of total squared singular value mass.
- Slow singular value decay (no sharp elbow) is typical of broad-vocabulary document collections.

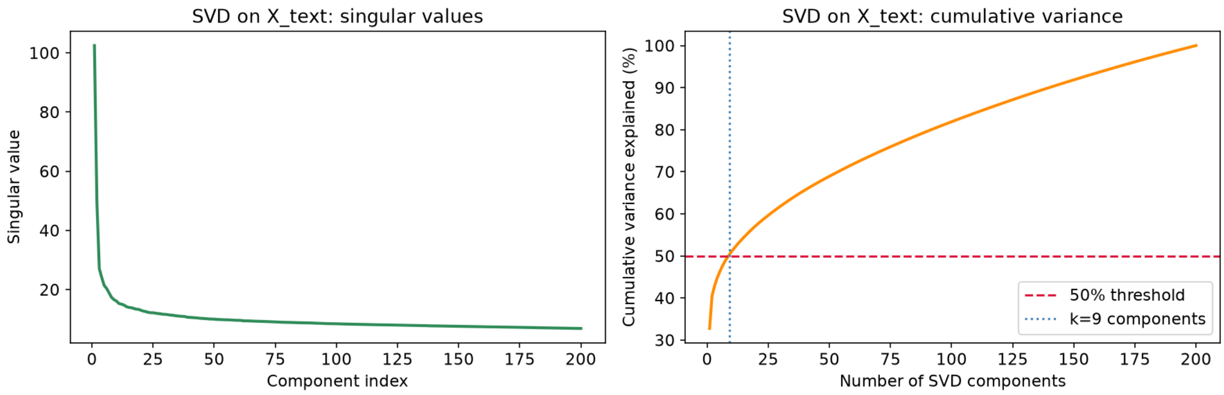

Figure: TruncatedSVD on X_text: singular value spectrum (left) and cumulative variance (right).

Summary

Matrix	Original dims	Reduced dims	Criterion	Method
X_numeric	26	12	>= 90% variance	PCA
X_text	5,000	201	>= 50% variance	TruncatedSVD
X_categorical	5,000	50 (fixed)	downstream use	TruncatedSVD

5. Feature Matrix Dimensions

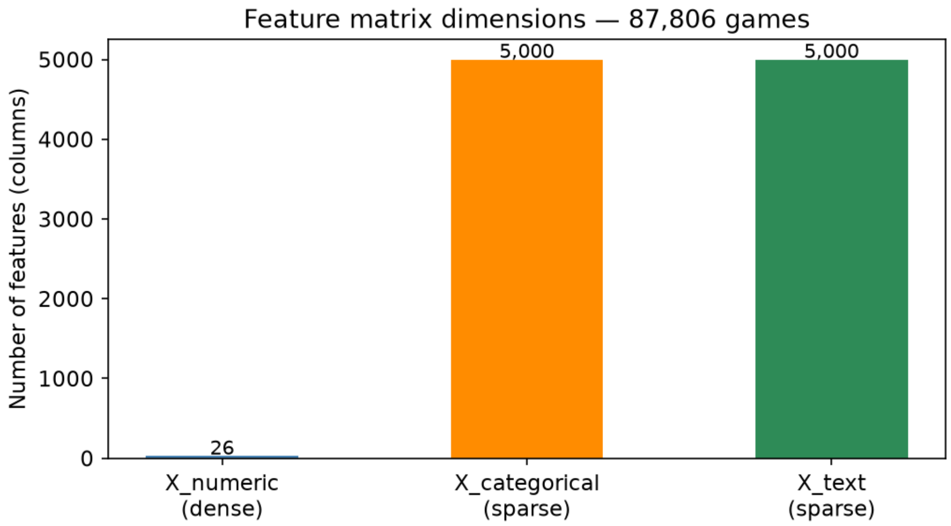

Figure: Feature column count per matrix. X_numeric is dense; X_categorical and X_text are sparse.

6. Reproducible Pipeline

```
# From project root
python src/feature_engineering.py
```

Artifacts: X_numeric.npz, X_categorical.npz, X_text.npz, models.pkl, metrics.json

All outputs deterministic (random_state=42).

7. Ethics and Access Note

- Provenance: Feature matrices derived from steam_v1.parquet. No review text or PII.
- Authorization: CC0 license; unrestricted academic use.
- Bias note: X_numeric includes popularity counts introducing popularity bias (addressed in W10).